

5

4

3

2

1

W101-001 REV A

NOTES:

1. INTERPRET DRAWING PER ASME Y14.100.

2. DIMENSIONS ARE IN INCHES. LENGTH TOLERANCE ±0.5 IN.

3. WIRE PER WIRE LIST. TERMINATE CONTACTS PER CONNECTOR MANUFACTURER'S INSTRUCTIONS.

4. 100% CONTINUITY AND ISOLATION TEST PER WIRE LIST.

5. ASSEMBLY VIEW AND CONNECTOR PINOUTS ON SHEET 2. WIRING DIAGRAM ON SHEET 3. WIRE LIST, WIRE GROUPS, SPLICES AND LABEL SCHEDULE ON SHEET 4.

6. DIMENSIONS PER IPC-D-620: HARNESS DIMENSIONS, INCLUDING BREAKOUT AND SPLICE LOCATIONS, ARE MEASURED FROM DATUM A, THE P1 CONNECTOR FACE. HARNESS LENGTH IS MEASURED FROM DATUM A TO THE FINAL TERMINATION. BREAKOUT LENGTHS ARE MEASURED FROM THE APPROXIMATE CENTERLINE OF THE HARNESS AT THE BREAKOUT TO THE FINAL TERMINATION.

7. CONTACTS (M39029/56-351, M39029/58-363) ARE SUPPLIED WITH CONNECTORS (FIND NOS. 1, 2); DO NOT ORDER SEPARATELY.

8. CONNECTOR PINOUTS SHOW THE FRONT (ENGAGING) FACE OF THE CONNECTOR CALLED OUT: INSERT ARRANGEMENT PER MIL-STD-1560, SOCKET INSERTS AS THE MIRROR IMAGE OF THE PIN INSERT. FILLED CAVITIES ARE WIRED; OPEN CAVITIES ARE NC. MASTER KEYWAY NOT SHOWN.

9. NC (NO CONNECTION) CONTACT POSITIONS: P1: N, P, R, S, T, U; P2: N, P, R, S, T, U.

10. TWIST THE WIRES OF EACH TWISTED GROUP TOGETHER OVER THEIR FULL LENGTH (SEE WIRE GROUPS TABLE).

11. CALCULATED BUNDLE DIAMETER AT CONNECTORS (REF): P1 0.257 IN, P2 0.257 IN.

12. TERMINATE SHIELDS PER WIRE GROUPS TABLE USING FIND NO. 13. INSULATE FLOATING SHIELD ENDS.

13. INSTALL SPLICE (FIND NO. 14) AT LOCATION SHOWN; SEE SPLICE TABLE.

14. RECOVER HEATSHRINK BOOT (FIND NO. 4) OVER BACKSHELL CABLE CLAMP.

15. INSTALL IDENTIFICATION LABEL (FIND NO. 5) ADJACENT TO CONNECTOR; MARK PER LABEL SCHEDULE.

16. INSTALL WIRE MARKER (FIND NO. 15) AT EACH END OF EACH WIRE; MARK WITH WIRE ID.

REVISIONS

ZONE

REV

DESCRIPTION

DATE

APPROVED

A

INITIAL RELEASE

2026-10-08

J. SMITH

15

26

TMS-SCE-1/8-2.0-9

MARKER, WIRE, 1/8 IN

16

14

1

M81824/1-1

SPLICE, CRIMP, ENVIRONMENT RESISTANT

13

13

2

M83519/2-3

SHIELD TERMINATION, SOLDER SLEEVE

12

12

48 IN

M22759/16-20-5

WIRE, 20 AWG, GRN

11

54 IN

M22759/16-22-0

WIRE, 22 AWG, BLK

10

48 IN

M22759/16-22-6

WIRE, 22 AWG, BLU

9

AR

M22759/16-22-9

WIRE, 22 AWG, WHT

8

48 IN

M22759/16-20-0

WIRE, 20 AWG, BLK

7

48 IN

M22759/16-20-2

WIRE, 20 AWG, RED

6

48 IN

M27500-22TG2T14

CABLE, SHIELDED TWISTED PAIR, 22 AWG

5

2

TMS-SCE-1/2-2.0-9

SLEEVE, IDENTIFICATION, 1/2 IN

15

4

2

202K153-25/225-0

BOOT, HEATSHRINK, STRAIGHT

14

3

2

M85049/38S15W

BACKSHELL, STRAIGHT, SHELL 15

2

1

D38999/26WD18PN

CONNECTOR, PLUG, 18 PIN

1

1

D38999/26WD18SN

CONNECTOR, PLUG, 18 SKT

FIND NO

QTY REQD

CAGE CODE

PART OR IDENTIFYING NO

NOMENCLATURE OR DESCRIPTION

NOTE

PARTS LIST

APPLICATION

NEXT ASSY

USED ON

A1000-001

FCC INST...

UNLESS OTHERWISE SPECIFIED:

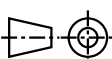
DIMENSIONS ARE IN INCHES

LENGTH TOLERANCE ±0.5

INTERPRET PER ASME Y14.5

DO NOT SCALE DRAWING

THIRD ANGLE PROJECTION



CONTRACT NO.

APPROVALS

DATE

DRAWN

K. KIPER

CHECKED

J. SMITH

ENGR

APPROVED

ACME AEROSPACE (EXAMPLE)

TITLE

CABLE ASSY, FCC TO PDU

SIZE

CAGE CODE

DWG NO.

REV

C

W101-001

A

SCALE

WEIGHT

SHEET

NONE

1 OF 4

5

4

3

2

1

5

4

3

2

1

W101-001 REV A

ASSEMBLY VIEW AND CONNECTOR PINOUTS (NOT TO SCALE)

C

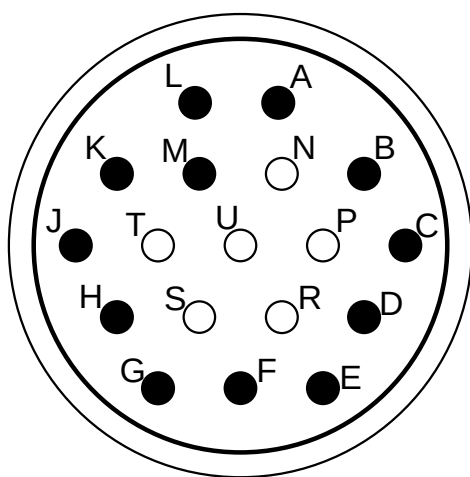
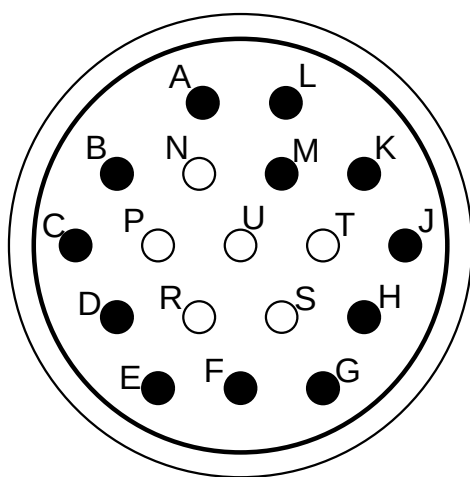
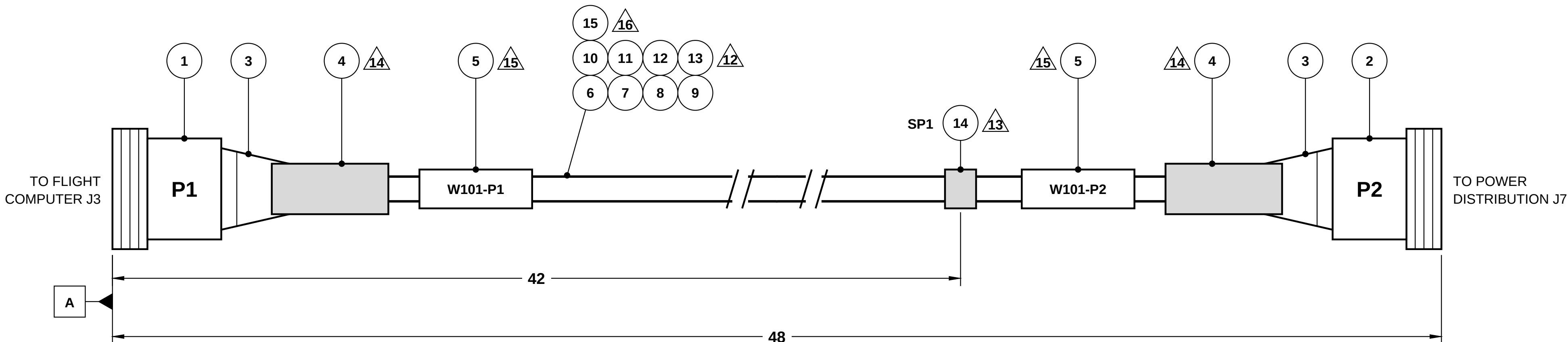
C

B

B

A

A



|                  |           |                            |                 |
|------------------|-----------|----------------------------|-----------------|
| SIZE<br><b>C</b> | CAGE CODE | DWG NO.<br><b>W101-001</b> | REV<br><b>A</b> |
| SCALE<br>NONE    | WEIGHT    | SHEET<br><b>2 OF 4</b>     |                 |

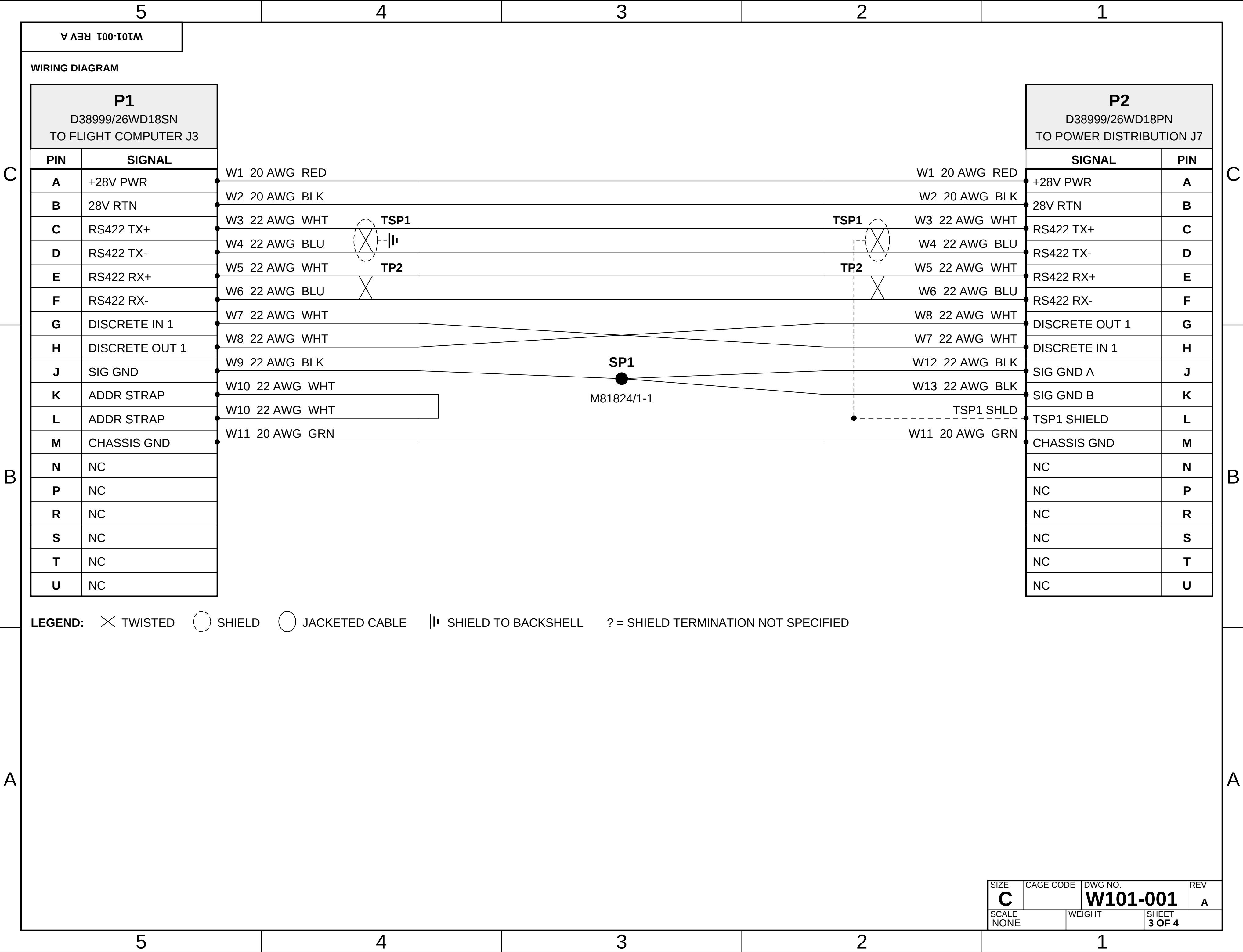
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|                         |                       |            |     |         |                              |                 |            |                          |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
|-------------------------|-----------------------|------------|-----|---------|------------------------------|-----------------|------------|--------------------------|----------------------|-----------------|--------------------------------|--|---|--|--|--|---|--|--|--|--|
| 5                       |                       |            |     |         | 4                            |                 |            |                          | 3                    |                 |                                |  | 2 |  |  |  | 1 |  |  |  |  |
| W101-001 REV A          |                       |            |     |         |                              |                 |            |                          |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
| WIRE LIST AND TABLES    |                       |            |     |         |                              |                 |            |                          |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
| WIRE LIST               |                       |            |     |         |                              |                 |            |                          |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
| WIRE                    | FROM                  | PIN        | TO  | PIN     | SIGNAL                       | AWG             | COLOR      | WIRE P/N                 | GROUP                | LENGTH (IN)     | NOTES                          |  |   |  |  |  |   |  |  |  |  |
| W1                      | P1                    | A          | P2  | A       | +28V PWR                     | 20              | RED        | M22759/16-20-2           |                      | 48              |                                |  |   |  |  |  |   |  |  |  |  |
| W2                      | P1                    | B          | P2  | B       | 28V RTN                      | 20              | BLK        | M22759/16-20-0           |                      | 48              |                                |  |   |  |  |  |   |  |  |  |  |
| W3                      | P1                    | C          | P2  | C       | RS422 TX+                    | 22              | WHT        | (M27500-22TG2T14)        | TSP1                 | 48              |                                |  |   |  |  |  |   |  |  |  |  |
| W4                      | P1                    | D          | P2  | D       | RS422 TX-                    | 22              | BLU        | (M27500-22TG2T14)        | TSP1                 | 48              |                                |  |   |  |  |  |   |  |  |  |  |
| W5                      | P1                    | E          | P2  | E       | RS422 RX+                    | 22              | WHT        | M22759/16-22-9           | TP2                  | 48              |                                |  |   |  |  |  |   |  |  |  |  |
| W6                      | P1                    | F          | P2  | F       | RS422 RX-                    | 22              | BLU        | M22759/16-22-6           | TP2                  | 48              |                                |  |   |  |  |  |   |  |  |  |  |
| W7                      | P1                    | G          | P2  | H       | DISCRETE IN 1                | 22              | WHT        | M22759/16-22-9           |                      | 48              |                                |  |   |  |  |  |   |  |  |  |  |
| W8                      | P1                    | H          | P2  | G       | DISCRETE OUT 1               | 22              | WHT        | M22759/16-22-9           |                      | 48              |                                |  |   |  |  |  |   |  |  |  |  |
| W9                      | P1                    | J          | SP1 |         | SIG GND                      | 22              | BLK        | M22759/16-22-0           |                      | 42              |                                |  |   |  |  |  |   |  |  |  |  |
| W10                     | P1                    | K          | P1  | L       | ADDR STRAP                   | 22              | WHT        | M22759/16-22-9           |                      | AR              | JUMPER IN P1                   |  |   |  |  |  |   |  |  |  |  |
| W11                     | P1                    | M          | P2  | M       | CHASSIS GND                  | 20              | GRN        | M22759/16-20-5           |                      | 48              |                                |  |   |  |  |  |   |  |  |  |  |
| W12                     | SP1                   |            | P2  | J       | SIG GND A                    | 22              | BLK        | M22759/16-22-0           |                      | 6               |                                |  |   |  |  |  |   |  |  |  |  |
| W13                     | SP1                   |            | P2  | K       | SIG GND B                    | 22              | BLK        | M22759/16-22-0           |                      | 6               |                                |  |   |  |  |  |   |  |  |  |  |
|                         |                       |            |     |         |                              |                 |            |                          |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
| WIRE GROUPS AND SHIELDS |                       |            |     |         |                              |                 |            |                          |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
| GROUP                   | TYPE                  |            |     |         | WIRES                        | CABLE P/N       | SHIELD P/N | SHIELD TERM              | SHIELD (FROM END)    | SHIELD (TO END) | NOTES                          |  |   |  |  |  |   |  |  |  |  |
| TSP1                    | SHIELDED TWISTED PAIR |            |     |         | W3, W4                       | M27500-22TG2T14 |            | M83519/2-3 (FIND NO. 13) | P1: BACKSHELL        | P2: PIN L       | RS422 TX                       |  |   |  |  |  |   |  |  |  |  |
| TP2                     | TWISTED PAIR          |            |     |         | W5, W6                       |                 |            |                          |                      |                 | RS422 RX; 10 TWISTS PER FOO... |  |   |  |  |  |   |  |  |  |  |
|                         |                       |            |     |         |                              |                 |            |                          |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
| SPLICES                 |                       |            |     |         |                              |                 |            |                          |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
| SPLICE                  | FIND NO               | SPLICE P/N |     |         | LOCATION                     |                 |            | WIRES                    | NOTES                |                 |                                |  |   |  |  |  |   |  |  |  |  |
| SP1                     | 14                    | M81824/1-1 |     |         | 42 IN FROM DATUM A (P1 FACE) |                 |            | W9, W12, W13             | SIGNAL GROUND SPLICE |                 |                                |  |   |  |  |  |   |  |  |  |  |
|                         |                       |            |     |         |                              |                 |            |                          |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
| LABEL SCHEDULE          |                       |            |     |         |                              |                 |            |                          |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
| FIND NO                 | LABEL P/N             |            |     | LEGEND  | LOCATION                     |                 |            | QTY                      |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
| 5                       | TMS-SCE-1/2-2.0-9     |            |     | W101-P1 | CABLE, ADJACENT TO P1        |                 |            | 1                        |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
| 5                       | TMS-SCE-1/2-2.0-9     |            |     | W101-P2 | CABLE, ADJACENT TO P2        |                 |            | 1                        |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
| 15                      | TMS-SCE-1/8-2.0-9     |            |     | W1      | WIRE W1, AT P1-A AND P2-A    |                 |            | 2                        |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
| 15                      | TMS-SCE-1/8-2.0-9     |            |     | W2      | WIRE W2, AT P1-B AND P2-B    |                 |            | 2                        |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
| 15                      | TMS-SCE-1/8-2.0-9     |            |     | W3      | WIRE W3, AT P1-C AND P2-C    |                 |            | 2                        |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
| 15                      | TMS-SCE-1/8-2.0-9     |            |     | W4      | WIRE W4, AT P1-D AND P2-D    |                 |            | 2                        |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
| 15                      | TMS-SCE-1/8-2.0-9     |            |     | W5      | WIRE W5, AT P1-E AND P2-E    |                 |            | 2                        |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
| 15                      | TMS-SCE-1/8-2.0-9     |            |     | W6      | WIRE W6, AT P1-F AND P2-F    |                 |            | 2                        |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
| 15                      | TMS-SCE-1/8-2.0-9     |            |     | W7      | WIRE W7, AT P1-G AND P2-H    |                 |            | 2                        |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
| 15                      | TMS-SCE-1/8-2.0-9     |            |     | W8      | WIRE W8, AT P1-H AND P2-G    |                 |            | 2                        |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
| 15                      | TMS-SCE-1/8-2.0-9     |            |     | W9      | WIRE W9, AT P1-J AND SP1     |                 |            | 2                        |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
| 15                      | TMS-SCE-1/8-2.0-9     |            |     | W10     | WIRE W10, AT P1-K AND P1-L   |                 |            | 2                        |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
| 15                      | TMS-SCE-1/8-2.0-9     |            |     | W11     | WIRE W11, AT P1-M AND P2-M   |                 |            | 2                        |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
| 15                      | TMS-SCE-1/8-2.0-9     |            |     | W12     | WIRE W12, AT SP1 AND P2-J    |                 |            | 2                        |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
| 15                      | TMS-SCE-1/8-2.0-9     |            |     | W13     | WIRE W13, AT SP1 AND P2-K    |                 |            | 2                        |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
|                         |                       |            |     |         |                              |                 |            |                          |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
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|                         |                       |            |     |         |                              |                 |            |                          |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
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|                         |                       |            |     |         |                              |                 |            |                          |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
|                         |                       |            |     |         |                              |                 |            |                          |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
|                         |                       |            |     |         |                              |                 |            |                          |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
|                         |                       |            |     |         |                              |                 |            |                          |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
|                         |                       |            |     |         |                              |                 |            |                          |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
|                         |                       |            |     |         |                              |                 |            |                          |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
|                         |                       |            |     |         |                              |                 |            |                          |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
|                         |                       |            |     |         |                              |                 |            |                          |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
|                         |                       |            |     |         |                              |                 |            |                          |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
|                         |                       |            |     |         |                              |                 |            |                          |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
|                         |                       |            |     |         |                              |                 |            |                          |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
|                         |                       |            |     |         |                              |                 |            |                          |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
|                         |                       |            |     |         |                              |                 |            |                          |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
|                         |                       |            |     |         |                              |                 |            |                          |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
|                         |                       |            |     |         |                              |                 |            |                          |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
|                         |                       |            |     |         |                              |                 |            |                          |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
|                         |                       |            |     |         |                              |                 |            |                          |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
|                         |                       |            |     |         |                              |                 |            |                          |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
|                         |                       |            |     |         |                              |                 |            |                          |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
|                         |                       |            |     |         |                              |                 |            |                          |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
|                         |                       |            |     |         |                              |                 |            |                          |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
|                         |                       |            |     |         |                              |                 |            |                          |                      |                 |                                |  |   |  |  |  |   |  |  |  |  |
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